An Image Retrieval System Based on Colors and Shapes of Objects

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Abstract

This paper proposes a context-based image retrieval method, called color-shape based method (CSBM), based on the colors and shapes of objects in an image. The proposed method characterizes the shape of an object by the object's area and the intercepted lengths obtained by intercepting the object perimeter by eight lines with different orientations passing through the object center. The experimental results show that CSBM provides a better performance than FCH and CCH. Besides, it is insensitive to translation, rotation, distortion, scale, and hue variations, but impressionable to contrast and noise variations.

1 Introduction

A tremendous number of images have been generated everyday in various applications. Hence, it is necessary for developing an efficient image retrieval technique to take care of images. Context-based image retrieval has been studied for more than two decades, which generally works with features including color, texture, and shape [5]. Color histogram is one of the most commonly adopted features for designing image retrieval systems [7-12]. The advantages of color histogram are simple in operation and easy for calculation. Additionally, the feature is less affected by the rotation and translation variations of the objects in an image. In order to improve the efficacy of the conventional color histogram (CCH) [17], a fuzzy-based technique, namely fuzzy color histogram (FCH) [13], is proposed for color histogram construction. The method considers the degree of color similarity for each pixel to be associated with all the histogram bins using fuzzy memberships, in which the memberships are evaluated based on the fuzzy c-means algorithm [14-16]. It is demonstrated to be more robust than the CCH in dealing with quantization errors and changes in light intensity.

However, color histogram has a significant drawback that it can delineate only the global properties of an image because the local information for individual objects is simply merged and diluted. For example, a color histogram cannot express the properties such as shapes and sizes of objects in an image. To overcome the problem above, this paper proposes an image retrieval method based on the features including the colors as well as areas of the objects, and the intercepted lengths obtained by intercepting the object perimeter by eight lines with different orientations passing through the object center. The intercepted lengths are valuable in discriminating shapes of the objects and are immune to translation and rotation variations. In addition, scaling invariance is achieved by normalizations that the area of each object is divided by the image size and the intercepted lengths by the sum of width and height of the image. We call this method color-shape based method (CSBM).

In CSBM, after inputting a query image I_q, the features of I_q are compared with those of each database image. The database images, most similar to I_q, are delivered and ranked according to their similarities to I_q. The experimental results show that this method can efficiently distinguish two images by the colors, areas, and shapes of the objects on both images. Its retrieval accuracy is very robust and less affected by translation, rotation, and scaling variations

2 Related Works

Color histogram has been widely used as a feature for image retrieval application [17]. It first classifies colors of the database images as k clusters, known as color bins, using K-mean algorithm [18], in which the representative of each cluster is obtained by calculating its centroid. Then, the color histogram of an image can be obtained by assigning each pixel to a color bin with smallest distance. By observing the distribution of the color histogram, one can distinguish how similar two images are. The measure of similarity between two images is defined as:

$$s = \sum_{j=1}^{k} \min(H_j(I), H_j(I')) \Big/ \sum_{j=1}^{k} H_j(I')$$

where $H_j(I)$ and $H_j(I')$ indicate the j-th bin of k color histograms for image I and I'. The advantages for using color histogram as a feature are that it is very simple in operation and easy for calculation. Additionally, it is immune to shift, rotation, and scaling operations [7-12]. Although it has the aforementioned advantages, there are still several accompanied drawbacks, including: (1) the feature can be determined only by the number of pixels in each color bin; (2) the number of color bins should be large enough to have better retrieval efficiency, which in turn needs more memory space and computation time; and (3) the feature only accounts for global rather than local characteristics in an image; hence it cannot be used for local image retrieval.

Since CCH considers neither color similarity across different bins nor color dissimilarity in the same bin, it is sensitive to noisy interference coming from illumination changes and quantization errors. Hence it motivated Han and Ma [13] to develop an image retrieval system based on the fuzzy color histogram (FCH) which associates the color similarity of each pixel to all the color bins through a fuzzy-set membership function. The FCH, $F(I) = [f_1, f_2, \dots, f_n]$, of an image I consisting N pixels can be defined as:

$$f_i = \sum_{j=1}^{N} \mu_{ij} P_j = \frac{1}{N} \sum_{j=1}^{N} \mu_{ij}$$

where N is the number of color bins in the FCH, P_j represents the probability of a pixel j selected from image I, and μ_{ij} ($0 \leqq \mu_{ij} \leqq 1$) indicates the membership degree of the j-th pixel to the i-th color bin.

To speed up the computation, fine uniform quantization is performed in the RGB color space by mapping all the pixel colors into n' color histogram bins, that n' must be large enough so that the color difference between two adjacent bins is significantly small. Then, the n' colors are transformed from RGB to CIELAB color space. Finally, it classifies the n' colors in CIELAB color space into n clusters using the FCM clustering technique that each cluster represents an FCH bin.

During image retrieval, the FCH of a query image is extracted using the above approach, and then the system compares the FCH with the FCHs of all the database images. Dissimilarity between two images with FCHs represented as F_Q and F_T which indicate query and database image, respectively, can be measured using the Euclidean distance, as defined below:

$$d^2(F_Q, F_T) = [F_Q - F_T]^{\mathrm{T}} [F_Q - F_T]$$

Fuzzy C-Means (FCM) [14-16] clustering algorithm is used to compute the FCH. It classifies the n' fine colors into n coarse colors and calculates the membership matrix at the same time. The optimal solution is obtained by minimizing an objective function J_m, which is the weighted sum of squared errors within each group:

$$J_m(U, V; X) = \sum_{k=1}^{n} \sum_{i=1}^{c} \mu_{ik}^{m} \left\| x_k - v_i \right\|_A^2$$

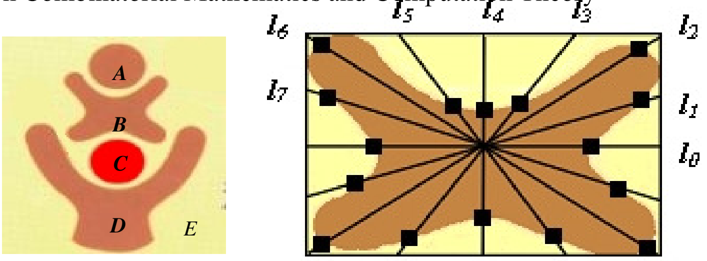
The algorithm stops when J_m is less then a given threshold ε. Here, $V = [v_1, v_2, \dots, v_c]^T$ represents a vector of unknown cluster prototypes, and the weighting exponent m controls the degree of membership shared by c clusters. The value of μ_{ik} indicates the degree of membership for a data point x_k in set $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$ related to i-th cluster. The membership for a data point to any cluster can be delineated by a matrix $\mathbf{U} = [\mu_{ik}]$, $\mu_{ik} \in [0,1]$, which satisfies the following constraints:

$$\sum_{k=1}^{n} \mu_{ik} = 1, \qquad 1 \le k \le n \text{ and } x_i \in X$$

The drawbacks of the FCH are that the feature can only be used for accounting the global characteristics and it needs a great computation time for calculating the memberships.

3 Color-Shape Based Method

In a true color image, a pixel is generally represented with a 24-bit number, such that there are totally 2^{24} different possible colors. If the possible pixels' colors in a full color image are reduced to a few ones, the image usually becomes the image consisting of several large regions, each of which is made up of a set of pixels with the same color. In the real world, there exist many such images, i.e. trademark, cartoon, flag, traffic sign, and synthesized images. Content in this sort of images is usually composed of a group of objects. However, segmenting the objects contained in an image is a difficult task. Thus, this paper regards a region, formed with identical color pixels, as one object. Two similar images generally contain some bigger objects that have similar colors and shapes. Figure 1(a) shows an image containing five objects A, B, C, D and E. As shown in this figure, objects A and B are usually treated as two different objects because



(a) (b)

Figure 1: (a) An image and (b) The *8* perimeter intercepts of one object bounded by an MBR

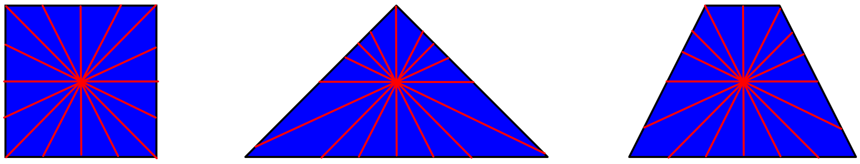


Figure 2: Objects having the same area but different shapes and PILs

they have different shapes even they have identical color. Objects ***A*** and ***C*** are also different because their colors are quite distinct although they have the same shapes and sizes.

3.1 Feature Extraction

The color-shape based method first classifies the pixels of all database images into ***K*** clusters using the K-mean clustering algorithm [18]. The mean value of each cluster is regarded as a representative color in a color palette, namely common color palette (*CCP*), for all images including database and query images. For an image I, a color-reduced image I' containing only K colors is generated by replacing each pixel in I with a closest color in *CCP*. Hence, I' has much less colors compared to I. The content of I' is generally comprised of a group of big objects, each with a uniform pixel color.

Every object has color, area, and shape. For an object O with its area greater than a threshold value Th_A, the features including color, area, and perimeter intercepted lengths (PIL) are computed and recorded; otherwise, O is treated as a noise and ignored. The area is featured as the number of pixels within O, whereas the shape as the lengths of **8** perimeter intercepts, $L=\{l_0, l_1, \ldots, l_7\}$, obtained by **8** lines intersecting the perimeter of O.

CSBM categorizes the directions from 0° to 360° into eight directions 0°, 22.5°, 45°, 67.5°, 90°, 112.5°, 135°, and 157.5° signified by θ_0, $\theta_1, \ldots$, and θ_7 respectively. The method gives a minimal bounding rectangle (MBR) containing O where the rectangle is oriented parallel to the X and Y axes, and it draws eight lines through the central pixel of ***O's*** MBR where the included angles between the eight lines and the X-axis are θ_0, $\theta_1, \ldots$, and θ_7 respectively. Figure 1 shows the eight lines. Let L_i be the i-th one of the eight lines, and P_1, P_2, …, P_e be the intersections of L_i and the boundary of O. We define l_i, the maximal distance between any two of P_1, P_2, …, P_e, as the length of L_i. For object O, CSBM writes down its color, area, and PILs $L=\{l_0, l_1, \ldots, l_7\}$ into database.

The shape of an identified object is very crucial for context-based approach to measure the similarity of two images. The difference between the PILs of two objects is demonstrated to be a good indicator of shape variation. In case that they are identical in areas but different in shape, their PILs are generally different. As shown in Figure 2, although the areas of all three objects are very close, their shapes are greatly different; hence their PILs are quite different too. To achieve rotation invariant, two objects are compared in *8* different orientations that the smallest Euclidean distance is used as an indicator for shape differentiation.

When a tool, like a scanner, is used to input an image, the image may be enlarged or shrunk because of different scanner resolution setups. We call this phenomenon scale variation. Let I be an image of $H\times W$ pixels. To remedy the problem of scale variations, this paper divides each value of PILs and the area of each object in I by $(H+W)$ and by $(H\times W)$, respectively. CSBM is hence able to resist the impact of scale variation.

3.2 Image Matching

Let O^q_{h1}, O^q_{h2}, …, $O^q_{hn^q_h}$ and O^d_{h1}, O^d_{h2}, …, $O^d_{hn^d_h}$ be all the objects with the h-th color of *CCP* respectively in the color-reduced images I'_q and I'_d of the query image I_q and one database image I_d. n^q_h and n^d_h are the number of objects, whose colors are the h-th color in *CCP*, in I'_q and I'_d. CSBM calculates the matching distance d_{ij} of two objects O^q_{hi} and O^d_{hj} via d^L_{ij} and d^A_{ij} as follows:

$$d_{ij} = d^L_{ij} + w\times d^A_{ij},$$

Algorithm *OBJ_Match*( ):

Input: O^q_{h1}, O^q_{h2}, …, $O^q_{hn^q_h}$ and O^d_{h1}, O^d_{h2}, …, $O^d_{hn^d_h}$.

Output: MMD^h

$MMD^h_{0,0} = 0$;

for $i = 0$ **to** n^q_h $\quad MMD^h_{i,0} = Penalty(O^q_{hi}) + MMD^h_{i-1,0}$;

for $j = 0$ **to** n^d_h $\quad MMD^h_{0,j} = Penalty(O^d_{hj}) + MMD^h_{0,j-1}$;

for $i = 1$ **to** n^q_h

for $j = 2$ **to** n^d_h

$MMD^h_{i,j} = min(MMD^h_{i-1,j} + Penalty(O^q_{hi}), MMD^h_{i,j-1} + Penalty(O^d_{hj}), MMD^h_{i,-1j-1} + d_{i,j})$

where w is a given constant; d^L_{ij} and d^A_{ij} are respectively the matching distances of PILs and areas of O^q_{hi} and O^d_{hj}. Assume that (l^q_0, l^q_1, …, l^q_7) and (l^d_0, l^d_1, …, l^d_7) are the PILs, and A^q_i as well as A^d_j are the areas of O^q_{hi} and O^d_{hj}. The color-shape based method defines d^L_{ij} and d^A_{ij} as follows, where z is a given constants.

$$d^L_{ij} = \sqrt[z]{\min_{r=0}^{7}(\sum_{s=0}^{7}\left|l^q_s - l^d_{(r+s) \bmod 8}\right|^z)}\text{, and}$$

$$d^A_{ij} = \left|A^q_i - A^d_j\right|.$$

CSBM adopts a dynamic programming algorithm ***OBJ_Match***( ) to calculate the minimal matching distance between two sets of objects O^q_{h1}, O^q_{h2}, …, $O^q_{hn^q_h}$ and O^d_{h1}, O^d_{h2}, …, $O^d_{hn^d_h}$. In this algorithm, a two-dimension array MMD^h, consisting of $n^q_h \times n^d_h$, is used to write down the minimal matching distance of O^q_{h1}, O^q_{h2}, …, $O^q_{hn^q_h}$ and O^d_{h1}, O^d_{h2}, …, $O^d_{hn^d_h}$. Entry $MMD^h_{i,j}$ records the minimal matching distance between O^q_{h1}, O^q_{h2}, …, O^q_{hi} and O^d_{h1}, O^d_{h2}, …, O^d_{hj}; therefore, entry $MMD^h_{n^q_h,n^d_h}$ describes the final minimal matching distance of O^q_{h1}, O^q_{h2}, …, $O^q_{hn^q_h}$ and O^d_{h1}, O^d_{h2}, …, $O^d_{hn^d_h}$. In this algorithm, each object in O^q_{h1}, O^q_{h2}, …, $O^q_{hn^q_h}$ may map into at most one object in O^d_{h1}, O^d_{h2}, …, $O^d_{hn^d_h}$, and vice versa. For an object O which does not be matched, the algorithm gives a penalty function ***Penalty***(***O***) which is defined as the following:

$$Penalty(O) = \sqrt[z]{\sum_{r=0}^{7} l^z_r} + A_r \times w,$$

where (l_1, l_2, …, l_7) are the PILs and A_r is the area of O.

Algorithm *OBJ_Match* is a function to compute the minimal matching distance of the objects with the h-th color of *CCP* in I'_q and I'_d. Finally, the color-shape based method defines the image matching distance ***Dist*** of I_q and I_d as

$$Dist = \sum_{h=1}^{K} MMD^h_{n^q_h,n^d_h} .$$

3.3 System Evaluation

In this work, the average normalized modified retrieval rank (ANMRR) proposed by MPEG-7 [19-20] are used as a benchmark for system evaluation. The ANMRR not only reflects the recall rate and precision information of the retrieved images, but also shows the ranks of all the retrieved images. Before defining ANMRR, several terms must be defined beforehand.

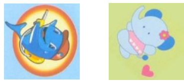
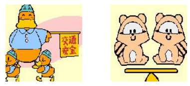
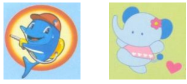
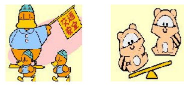
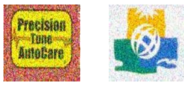
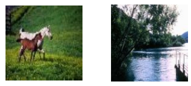
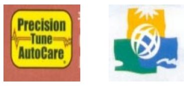
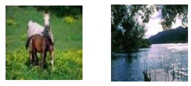
For each query q, the averaged rank (***AVR***) can be obtained by:

$$AVR(q) = \sum_{t=1}^{I(q)} R'(t)/I(q),$$

in which $I(q)$ indicates the number of images in the database which are most similar to the query image; $R(t)$ represents the rank of a returned image, which is defined as:

$$R'(t) = \begin{cases} R(t), \text{if } R(t) \le T \\ T+1, \text{if } R(t) > T \end{cases},$$

Table 1: Examples of classified query and database image pairs.

Category	Rotation	Scaling	Translation
Query Image			
Database Image			
Category	**Noise**	**Blurring**	**Natural Image**
Query Image			
Database Image			

where T is defined by $\min(4\times NG(q), 2\times GTM)$ with GTM representing $\max\{I(q)\}$ for each query, and t indicates the number of returned images which are most similar to the query image q.

For each query image, the value of AVR depends on $I(q)$; it is necessary to minimize the influence of $I(q)$ on ranking. Hence, the modified retrieval rank (MRR) can be defined as:

$$MRR(q) = AVR(q) - 0.5 - \frac{I(q)}{2}.$$

As shown in the above equation, the value of MRR depends on $I(q)$, which can be normalized to the value between 0 and 1. The normalized modified retrieval rank ($NMRR$) was defined as:

$$NMRR(q) = \frac{MRR(q)}{T + 0.5 - 0.5 \times I(q)}.$$

For a perfect retrieval $NMRR=0$, which indicates that the system can successfully retrieve all the images that are similar to the query image. If the system retrieves none of the images, then $NMRR=1$. Finally, by considering all the queries, the performance index can be obtained by calculating the averaged normalized modified retrieval rank (ANMRR), as shown below:

$$ANMRR = \sum_{q=1}^{Q} NMRR(q) \Big/ Q,$$

where Q is the total number of queries.

For example, if a system which is requested to return the most 15 similar images, then $T=15$. Furthermore, if there are 6 images retrieved by the system for each query image, then $I(q) = 6$. Suppose the most four similar images are ranked as 1, 4, 9, and 14, respectively, while the other two images are ranked outside the top 15, the values of $AVR(q)$ and $MRR(q)$ are 10 and 0.52, respectively.

4 Experiments and Discussions

This section investigates the performance of CSBM and compares it to those of FCH and CCH by two experiments. The first experiment is to test the capacity for recognizing images, while the second experiment explores its sensitivities to the **rotation, scaling, translation, noise, blurring, luminance, contrast, hue, and distortion** variations of images. According to our other experiments, when given the weighted parameters w and z to be 0.009 and 1.1 respectively, CSBM can give a better performance. Hence, the following experiments will specify both parameters to be above values.

The first experiment uses two sets of hybrid images $Set_D=\{I^d_1, I^d_2,\ldots, I^d_{500}\}$ and $Set_Q=\{I^q_1, I^q_2,\ldots, I^q_{500}\}$ as testing images**.** Each of the two image sets consists of 500 full color images including synthesized images and natural images. The images in Set_D are used as the database images, while those in Set_Q as the query images. In this experiment, each image pair (I^q_i, I^d_i) are considered to be similar images, that is, the user obtaining the correct answer, if I^d_i is delivered for given I^q_i as the query image. Table 1 shows a small part of images in both sets, which have different variations. Table 2 shows the performances of CSBM for CCP's size K (= 16 and 27) with three different area thresholds Th_A (= 5, 10, and 20).

As shown in Table 2, CSBM provides better performances than FCH and CCH whatever they are measured by ANMRR.

Table 2: The performances of CSBM, FCH, and CCH in the first experiment

Method	*K*	*Th_A*	*R'(t)=1*	*R'(t)=2*	*R'(t)= 5*	*ANMRR*
CSBM	16	*5*	405	432	448	0.163
		10	410	439	460	0.151
		20	397	421	440	0.182
	27	*5*	411	435	456	0.154
		10	397	427	452	0.176
		20	392	419	445	0.189
FCH	N/A		393	412	430	0.195
CCH	N/A		355	398	425	0.247

Table 3: Partial examples of trademark images and their geometric variations

Category	**Query image**	**Trans. & Rot.**	**Luminance**	**Contrast**	**Noise**	**Distortion**	**Scaling**	**Hue**
NO. 19								
NO. 56								
NO. 70								
NO. 89								

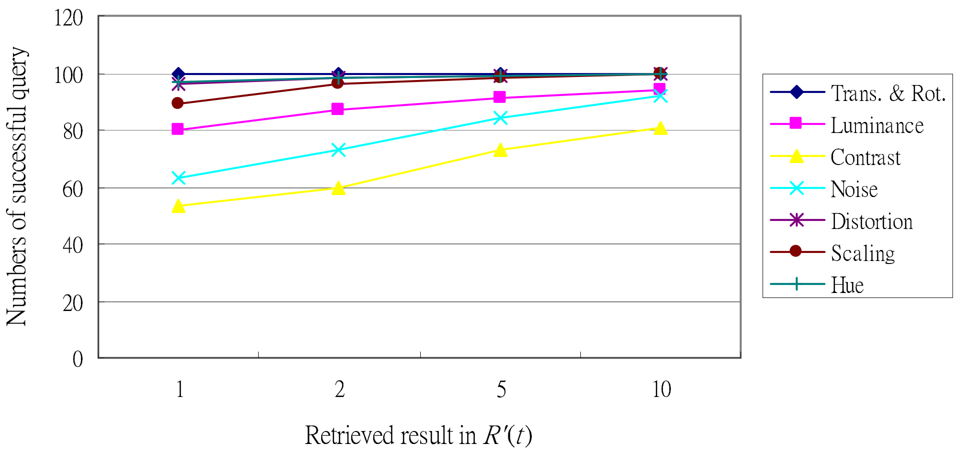


Figure 3: The performances of CSBM in different geometric variations

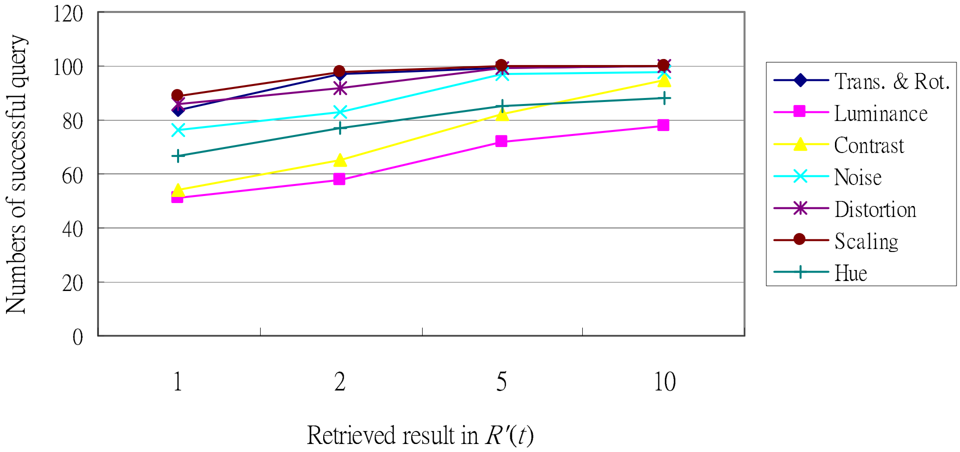


Figure 4: The performances of FCH in different geometric variations

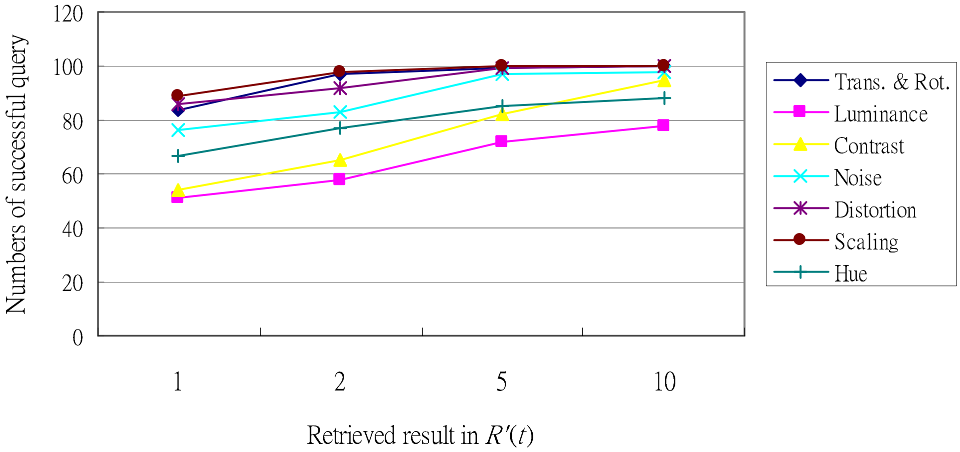


Figure 5: The performance of CCH in different geometric variations

In the second experiment, ***100*** full color images $I_{q,1}$, $I_{q,2}, \ldots, I_{q,100}$ are used as the query images. These ***100*** full color images make up an image set S_q. Besides, this paper employs the rotation, distortion, noise, scale, hue, luminance, and contrast functions in **Adobe Photoshop 7.0** to process each $I_{q,i}$, and then respectively generates the variant images $I_{r,i}$, $I_{d,i}$, $I_{n,i}$, $I_{s,i}$, $I_{h,i}$, $I_{l,i}$, and $I_{c,i}$. The group of images $I_{\alpha,1}, I_{\alpha,2}, \ldots, I_{\alpha,100}$ forms an image set S_α, which are produced by the same function, for $\alpha = r, d, n, s, h, l$, and c. Table 3 shows some images in S_q, S_r, S_d, S_n, S_s, S_h, S_l, and S_c respectively, where i is the order numbers of the images in S_q, S_r, S_d, S_n, S_s, S_h, S_l, and S_c. In this experiment, K and Th_A are given to be ***27*** and ***5***.

Figures 3-5 respectively show the performances of CSBM, FCH, and CCH obtained in this experiment. The experimental results demonstrate that CSBM is indifferent to translation, rotation, distortion, scale, and hue variations, but very sensitive to contrast and noise variations, and slightly susceptible to luminance variation. FCH and CCH can resist scale variation, but very susceptive to luminance, contrast variations and slightly to noise and distortion variations. Moreover, FCH is very vulnerable to translation and rotation variations and CCH to hue variation.

Since the same objects only with different luminance or color contrasts may be regarded as different objects, CSBM, FCH, and CCH cannot give a very good performance. Noises may generate various small isolated objects which will affect the minimal matching distance between two groups of objects with identical color; therefore, CSBM is sensitive to noise variation. On the whole, CSBM is more robust in resisting translation, rotation, luminance, distortion, hue variations, and more susceptible to noise variation than FCH and CCH.

5 Conclusions

This paper provides CSBM to help a user retrieve images from a huge image database. The method adopts the colors, areas, and PILs of objects to describe the properties of an image. The area and PILs of an object can effectively characterize the shape of the object. In this method, a dynamic programming algorithm is used to compute the similarity between two groups of objects based on their color, areas and PILs. The experimental results tell that CSBM obtains a better performance than FCH and CCH. Mainly, CSBM is more powerful in resisting translation, rotation, luminance, distortion, hue variations, and more vulnerable to noise variation than FCH and CCH.

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References

[1] W. Cai, D. Feng and R. Fulton. Content-Based Retrieval of Dynamic PET Functional Images. *IEEE Transactions on Information Technology in Biomedicine, Vol. 4 (2000) 152-158.*

[2] D. J. Foran, D. Comaniciu, P. Meer, and L. A. Goodell. Computer-Assisted Discrimination Among Malignant Lymphomas and Leukemia Using Immunophenotyping, *Intelligent Image Repositories, and Telemicroscopy. IEEE Transactions on Information Technology in Biomedicine., Vol. 4 (2000) 265-273.*

[3] P. H. Lewis, K. Martinez, F. S. Abas, M. F. A. Fauzi, S. C. Y. Chan, M. J. Addis, M. J. Boniface, P. Grimwood, A. Stevenson, C. Lahanier, and J. Stevenson. An Integrated Content and Metadata Based Retrieval System for Art. *IEEE Transactions on Image Processing., Vol. 13 (2004) 302-313.*

[4] A. Kushki, P. Androutsos, K. N. Plataniotis, and A. N. Venetsanopoulos. Retrieval of Images From Artistic Repositories Using a Decision Fusion Framework. IEEE *Transactions on Image Processing*, *Vol. 13 (2004) 277-292.*

[5] A. W. M. Smeulders, M. Worring, S. Santini, A. Gupta, and R. Jain. Content-Based Image Retrieval at the End of the Early Years. *IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 22 (Dec. 2000) 1349-1380.*

[6] A. Mojsilovic and B. E. Rogowitz. Semantic Metric for Image Library Exploration. *IEEE Trans. Multimedia, Vol. 6 (2004) 828-838.*

[7] R. Brunelli and O.`Mich. Histograms Analysis for Image Retrieval. *Pattern Recognition, Vol. 34 (2001) 1625-1637.*

[8] T. Chan and L. H .Chen. Fast Mapping Algorithm for Histogram to Binary Set Conversion. *Pattern Recognition Letter, Vol.21 (2000) 899-906.*

[9] T. Gevers. Robust Histogram Construction from Color Invariants. *The Eighth IEEE International Conference on Computer Vision, Vol. 1 (2001) 615-620.*

[10] D. Guillamet and J. Vitria. A Comparison of Global Versus Local Color Histograms for Object Recognition. *The 15th International Conference on Pattern Recognition, Vol. 2 (2000) 422-425.*

[11] J. Hafner. Efficient color histogram indexing for quadratic form distance functions. *IEEE Transactions on Pattern Analysis and Machine intelligence (1996) 729-736.*

[12] W. Lu and Y. P. Tan. A Color Histogram Based People Tracking System. *The IEEE International Symposium on Circuits and Systems, Vol. 2 (2001) 137-140.*

[13] J. Han and K. K. Ma Fuzzy color histogram and its use in color image retrieval. *IEEE Transactions on Image Processing, Vol. 11 (Aug. 2002) 944-952.*

[14] J. C. Bezdek. Pattern Recognition with Fuzzy Objective Function Algorithms. *Plenum Press, New York (1981).*

[15] J. C. Dunn. A fuzzy relative of the ISODATA process and its use in detecting compact, well separated clusters. *J. Cybernet 3 (1974) 32-57.*

[16] A. F. Sintas, J. M. Cadenas, and F. Martin. Membership functions in the fuzzy C-Means Algorithm. *Fuzzy sets and Systems, Vol. 101 (1999) 49-58.*

[17] M. J. Swain and D. H. Ballard. Color indexing. *International Journal of Computer Vision, Vol. 7, No. 1 (1991) 11-32.*

[18] M. C. Su and C. H. Chou. A Modified Version of the K-means Algorithm with a Distance Based on Cluster Symmetry. *IEEE Transactions on Pattern Analysis and Machine Intelligence, (2001) 674-680.*

[19] B. S. Manjunath, J. R. Ohm, V. V. Vasudevan, and A. Yamada Color and texture descriptors. *IEEE Transaction On Circuits and Systems for Video Technology, Vol. 11, No. 6 (June 2001) 703-715.*

[20] V. V. Vasudevan and B. S. Manjunath. Report on AHG of color and texture. *ISO/IEC/JTC1/SC29/WG11, Doc. M5560, Maui (Dec. 1999) 332.*